

FISA - Recipes for Quadratic form

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1 Scalar product?

Let $\mathbf{x}$ defined as $\begin{bmatrix} x_1 & x_2 & \dots & x_m \end{bmatrix}^T$ and $\mathbf{y}$ as $\begin{bmatrix} y_1 & y_2 & \dots & y_m \end{bmatrix}^T$. The scalar product $\langle \mathbf{x}, \mathbf{y} \rangle$ of two vectors $\mathbf{x}$ and $\mathbf{y}$ of dimensions $(m \times 1)$ is the scalar that is obtained by summing the products of the respective components in a given basis:

$$\langle \mathbf{x}, \mathbf{y} \rangle = x_1 y_1 + x_2 y_2 + \dots + x_m y_m = \mathbf{x}^T \mathbf{y} = \mathbf{y}^T \mathbf{x}$$

The norm of a vector can be defined as $\|\mathbf{x}\| = \sqrt{\mathbf{x}^T \mathbf{x}}$.

It is possible to show the Schwarz' inequality hold. $|\mathbf{x}^T \mathbf{y}| \leq \|\mathbf{x}\| \|\mathbf{y}\|$

2 Bilinear form?

A bilinear form in the variables x_i and y_j is the scalar

$$B = \sum_{i=1}^m \sum_{j=1}^n a_{ij} x_i y_j$$

which can be written in matrix form

$$B(x, y) = \mathbf{x}^T \mathbf{A} \mathbf{y} = \mathbf{y}^T \mathbf{A}^T \mathbf{x}$$

where $\mathbf{x} = \begin{bmatrix} x_1 & x_2 & \dots & x_m \end{bmatrix}^T$, $\mathbf{y} = \begin{bmatrix} y_1 & y_2 & \dots & y_n \end{bmatrix}^T$, and $\mathbf{A}$ is the $(m \times n)$ matrix of the coefficients a_{ij} representing the core of the form.

What is the size of B(x, y)?

$1 \times m \times m \times n \times n \times 1 = 1 \times 1$; i.e. a scalar

How can you write this $\mathbf{x}^T \mathbf{A} \mathbf{y} = \mathbf{y}^T \mathbf{A}^T \mathbf{x}$?

The Bilinear form is a scalar (1x1). So I can write $scalar = scalar^T$

HINTS: use $(\mathbf{A}\mathbf{B})^T = \mathbf{B}^T \mathbf{A}^T$

$$\mathbf{x}^T \mathbf{A} \mathbf{y} = (\mathbf{x}^T \mathbf{A} \mathbf{y})^T = (\mathbf{A} \mathbf{y})^T (\mathbf{x}^T)^T = \mathbf{y}^T \mathbf{A}^T \mathbf{x}$$

Given the bilinear form, the gradient of the form with respect to $\mathbf{x}$ is given by

$$\text{grad}_{\mathbf{x}} B(x, y) = \left(\frac{\partial B(\mathbf{x}, \mathbf{y})}{\partial \mathbf{x}} \right)^T = \mathbf{A} \mathbf{y}$$

But How?

$$\text{grad}_{\mathbf{x}} B(x, y) = \left(\frac{\partial B(\mathbf{x}, \mathbf{y})}{\partial \mathbf{x}} \right)^T = \left(\frac{\partial \mathbf{x}^T \mathbf{A} \mathbf{y}}{\partial \mathbf{x}} \right)^T = \left(\frac{\partial \mathbf{y}^T \mathbf{A}^T \mathbf{x}}{\partial \mathbf{x}} \right)^T = (\mathbf{y}^T \mathbf{A}^T)^T = \mathbf{A} \mathbf{y}$$

whereas the gradient of B with respect to y is given (same demonstration) by

$$\text{grad}_y B(x, y) = \left(\frac{\partial B(x, y)}{\partial y} \right)^T = \mathbf{A}^T x$$

3 Quadratic form?

A special case of bilinear form is the quadratic form

$$Q(x) = x^T \mathbf{A} x$$

Given the quadratic form with $\mathbf{A}$ symmetric ($A = A^T$), the gradient of the form with respect to x is given by

$$\text{grad}_x Q(x) = \left(\frac{\partial Q(x)}{\partial x} \right)^T = 2\mathbf{A}x$$

But How?

We therefore see that we must derive with respect to x an expression where x appears twice: once on the right, and once on the left. To visualize this, let's denote these two vectors in blue and red (this is just a play of color: whether it is written in red or blue, the vector x always represents the same thing!):

$$\frac{\partial}{\partial \mathbf{x}} (\mathbf{x}^T \mathbf{A} \mathbf{x}) = \frac{\partial}{\partial \mathbf{x}} (\mathbf{x}^T \mathbf{A} \mathbf{x}) + \frac{\partial}{\partial \mathbf{x}} (\mathbf{x}^T \mathbf{A} \mathbf{x})$$

The derivative with respect to the red x does not pose a problem: you just need to apply the little math reminder framed above:

$$\frac{\partial}{\partial \mathbf{x}} (\mathbf{x}^T \mathbf{A} \mathbf{x}) = \mathbf{A} \mathbf{x}$$

For the derivative with respect to the blue $\mathbf{x}$ we must manage to bring the blue $\mathbf{x}$ to the left. To do this, we use the fact that Quadratic form is... a scalar! We can therefore happily transpose a scalar: this does not change it!

$$\frac{\partial}{\partial \mathbf{x}} (\mathbf{x}^T \mathbf{A} \mathbf{x}) = \frac{\partial}{\partial \mathbf{x}} (\mathbf{x}^T \mathbf{A} \mathbf{x})^T$$

HINTS: use $(\mathbf{A}\mathbf{B})^T = \mathbf{B}^T \mathbf{A}^T$ and $A = A^T$ i.e. $\mathbf{A}$ is symmetric.

It leads to:

$$\frac{\partial}{\partial \mathbf{x}} (\mathbf{x}^T \mathbf{A} \mathbf{x})^T = \frac{\partial}{\partial \mathbf{x}} (\mathbf{A} \mathbf{x})^T (\mathbf{x}^T)^T = \frac{\partial}{\partial \mathbf{x}} (\mathbf{x})^T \mathbf{A}^T \mathbf{x} = \mathbf{A} \mathbf{x}$$

This leads to:

$$\frac{\partial}{\partial \mathbf{x}} (\mathbf{x}^T \mathbf{A} \mathbf{x}) = \mathbf{A} \mathbf{x} + \mathbf{A} \mathbf{x} = 2\mathbf{A} \mathbf{x}$$

4 Check with Python?

Consider a matrix of quadratic form given hereafter

$$A = \begin{bmatrix} 3 & 2 & 0 \\ 2 & -1 & 4 \\ 0 & 4 & -2 \end{bmatrix}$$

construct the quadratic form $\mathbf{x}^T A \mathbf{x}$.

$$\begin{aligned} \mathbf{x}^T A \mathbf{x} &= \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} 3 & 2 & 0 \\ 2 & -1 & 4 \\ 0 & 4 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \\ &= \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} 3x_1 + 2x_2 \\ 2x_1 - x_2 + 4x_3 \\ 4x_2 - 2x_3 \end{bmatrix} \\ &= x_1(3x_1 + 2x_2) + x_2(2x_1 - x_2 + 4x_3) + x_3(4x_2 - 2x_3) \\ &= 3x_1^2 + 4x_1x_2 - x_2^2 + 8x_2x_3 - 2x_3^2 \end{aligned}$$

Fortunately, there is an easier way to calculate quadratic form. Notice that coefficients of x_i^2 is on the principal diagonal and coefficients of $x_i x_j$ are be split evenly been (i, j) - and (j, i) - entries in A .

Consider another example,

$$A = \begin{bmatrix} 3 & 2 & 0 & 5 \\ 2 & -1 & 4 & -3 \\ 0 & 4 & -2 & -4 \\ 5 & -3 & -4 & 7 \end{bmatrix}$$

All x_i^2 's terms are

$$3x_1^2 - x_2^2 - 2x_3^2 + 7x_4^2$$

whose coefficients are from principal diagonal.

All $x_i x_j$'s terms are

$$4x_1x_2 + 0x_1x_3 + 10x_1x_4 + 8x_2x_3 - 6x_2x_4 - 8x_3x_4$$

Add up together then quadratic form is

$$3x_1^2 - x_2^2 - 2x_3^2 + 7x_4^2 + 4x_1x_2 + 0x_1x_3 + 10x_1x_4 + 8x_2x_3 - 6x_2x_4 - 8x_3x_4$$

Let's verify in SymPy.

```
1 import sympy as sy
2
3 x1, x2, x3, x4 = sy.symbols('x1 x2 x3 x4')
4
5 A = sy.Matrix([
6     [3, 2, 0, 5],
7     [2, -1, 4, -3],
8     [0, 4, -2, -4],
9     [5, -3, -4, 7]
10 ])
11
12 x = sy.Matrix([x1, x2, x3, x4])
13
14 expr = x.T * A * x
15 result = sy.expand(expr[0]) # extract scalar from 1x1 matrix
16
17 print(result)
```

Listing 1: Python example

The result is exactly the same as we derived

$$3x_1^2 - x_2^2 - 2x_3^2 + 7x_4^2 + 4x_1x_2 + 0x_1x_3 + 10x_1x_4 + 8x_2x_3 - 6x_2x_4 - 8x_3x_4$$

What you can also do is recover A from this development ...since your result is a quadratic form

$$f(x) = x^T A x$$

you can recover A from the Hessian of the expanded expression.
For any quadratic form:

$$\nabla^2 f(x) = 2A$$

when A is symmetric (and your matrix is symmetric), so:

$$A = \frac{1}{2} \text{Hessian}(f)$$

```
1 # Variables vector
2 vars = [x1, x2, x3, x4]
3
4 # Hessian matrix
5 H = sy.hessian(result, vars)
6
7 # Recover A
8 A = H / 2
9
10 print("Hessian:")
11 print(H)
12
13 print("Recovered A:")
14 print(A)
```

Listing 2: Python example

5 References

- Lectures of Linear Algebra <https://github.com/weijie-chen/Linear-Algebra-With-Python/>
- Linear Algebra and Its Applications by Gilbert Strang
- Linear Algebra and Its Applications by David Lay
- Introduction to Linear Algebra With Applications by DeFranza and Gagliardi
- Linear Algebra With Applications by Gareth Williams